

ELEC5305 Project Proposal

Evaluating the Impact of Lossy Audio Compression on Automatic Speech Recognition Performance

1. Project Title

Evaluating the Impact of Lossy Audio Compression on Automatic Speech Recognition Performance

2. Student Information

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3. Project Overview

Automatic speech recognition (ASR) is increasingly used in applications where recorded speech must be stored or transmitted under limited bandwidth or storage capacity. Lossy audio codecs such as MP3 and Opus can greatly reduce audio data size, but they achieve this by removing or modifying information from the original signal. Although compressed speech may remain understandable to a human listener, the acoustic information used by an ASR system may respond differently to these distortions.

This project investigates the robustness of ASR to lossy audio compression at different bitrates. Speech recordings will be compressed using MP3 and Opus and compared with an uncompressed WAV baseline. All signals will then be processed using the same pretrained Wav2Vec2 speech recognition model. The main research question is: How robust is automatic speech recognition to lossy audio compression at different bitrates, and at what compression level does recognition performance begin to degrade significantly?

Word Error Rate (WER) will measure recognition accuracy, while bitrate, file size and compression ratio will measure compression efficiency. The project therefore aims to identify practical trade-offs between reduced audio data size and ASR performance.

4. Background and Motivation

Lossy audio codecs exploit properties of human perception to reduce the amount of information required to represent an audio signal. However, information that is less important to human perception is not necessarily irrelevant to machine-based speech recognition. This creates a practical engineering question for systems that transmit speech to cloud or embedded ASR services: how far can the audio bitrate be reduced before recognition reliability is meaningfully affected?

Previous work has shown that MP3 compression changes both the spectral and temporal characteristics of speech. Borsky et al. reported that low-bitrate MP3 coding can introduce strong spectral distortions and investigated their effect on ASR performance [1]. Opus has also been studied in bandwidth-constrained recognition systems; Drude et al. examined the relationship between Opus compression, bitrate allocation and word error rate in far-field ASR [2].

Modern ASR systems differ significantly from many systems used in earlier compression studies. Wav2Vec2 uses self-supervised representation learning and has demonstrated strong speech-recognition performance [3]. LibriSpeech provides a standard 16 kHz English speech corpus with reference transcriptions for controlled ASR evaluation [4]. This motivates testing whether a modern pretrained model remains robust under conventional MP3 and Opus compression.

The topic was selected because it combines digital audio signal processing with a concrete downstream application. The ASR model is held fixed while the input codec and bitrate are varied, producing a controlled experimental design that is technically meaningful but achievable within the semester.

5. Proposed Methodology

The project will be implemented in Python under Linux/WSL. PyTorch and TorchAudio will be used for the Wav2Vec2 ASR pipeline, FFmpeg will perform controlled MP3 and Opus encoding, and JiWER will calculate word error rate. The main dataset will be LibriSpeech test-clean. A fixed set of utterances will be selected using a reproducible random seed so that every codec condition is evaluated on exactly the same speech samples.

For each utterance, the original waveform will first be processed directly by Wav2Vec2 to establish the uncompressed WAV baseline. The same waveform will then be encoded at each selected bitrate, decoded back to an audio waveform, and processed by the identical ASR model. Initial conditions are MP3 at 128, 64 and 32 kbps, and Opus at 64, 32 and 16 kbps. If these settings cause little degradation, lower bitrate conditions will be introduced to identify a practical degradation threshold.

Recognition performance will be measured using $WER = (S + D + I) / N$, where S, D and I are substitutions, deletions and insertions, and N is the number of words in the reference transcription. Compression efficiency will be measured using $CR = \text{Original WAV Size} / \text{Compressed File Size}$. The main quantitative comparisons will therefore be WER versus bitrate and WER versus compression ratio.

To provide signal-processing interpretation rather than only a benchmark, representative waveforms and spectrograms will be examined across compression levels. Recognition errors will also be inspected to determine whether aggressive compression is associated with identifiable changes in substitutions, deletions or insertions.

Initial experimental conditions: WAV (uncompressed baseline); MP3 at 128, 64 and 32 kbps; Opus at 64, 32 and 16 kbps.

6. Expected Outcomes

The project is expected to deliver a reproducible working pipeline that automatically compresses speech, performs ASR, calculates WER, measures compression ratio and exports results for analysis. Final outputs will include comparison tables, WER-versus-bitrate plots, WER-versus-compression-ratio plots, selected spectrogram comparisons and qualitative examples of recognition errors.

A pilot implementation has already been completed. Preliminary evaluation on 100 randomly selected LibriSpeech test-clean utterances indicates that Wav2Vec2 remains relatively robust across the compression levels tested so far. These differences are small and are not treated as final conclusions. A larger evaluation set and more aggressive bitrate conditions will be used to determine whether a meaningful performance degradation threshold exists.

All working code, setup instructions, selected results and project documentation will be maintained in the public GitHub repository and GitHub Project Site.

7. Timeline (Weeks 1-13)

Weeks	Planned Work
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Weeks	Planned Work
1-2	Project topic selection and research question development
3-5	Literature review, dataset preparation and baseline testing
6-9	MP3/Opus pipeline implementation and initial experiments
10-11	Larger-scale testing, lower-bitrate experiments and signal/error analysis
12-13	Final evaluation, report, demonstration and GitHub documentation

8. References

- [1] M. Borsky, P. Mizera, P. Pollak, and J. Nouza, "Dithering techniques in automatic recognition of speech corrupted by MP3 compression: Analysis, solutions and experiments," *Speech Communication*, vol. 86, pp. 75-84, 2017.
- [2] L. Drude, J. Heymann, A. Schwarz, and J.-M. Valin, "Multi-Channel Opus Compression for Far-Field Automatic Speech Recognition with a Fixed Bitrate Budget," *Proc. Interspeech 2021*, pp. 1669-1673, 2021.
- [3] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, "wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations," *Advances in Neural Information Processing Systems*, vol. 33, 2020.
- [4] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, "LibriSpeech: An ASR Corpus Based on Public Domain Audio Books," *Proc. IEEE ICASSP*, pp. 5206-5210, 2015.

9. Appendix - Preliminary Results

Pilot results from 100 randomly selected LibriSpeech test-clean utterances are shown below. These results are preliminary and are included to demonstrate feasibility rather than to support final conclusions.

Format	Bitrate	Overall WER	Avg. Compression Ratio
WAV	Uncompressed	3.41%	1.00x
MP3	128 kbps	3.41%	1.95x
MP3	64 kbps	3.11%	3.90x
MP3	32 kbps	3.16%	7.78x
Opus	64 kbps	3.52%	3.60x
Opus	32 kbps	3.21%	8.08x
Opus	16 kbps	3.31%	15.91x

The current pilot suggests that recognition remains close to the uncompressed baseline over the tested range. The apparent small improvements under some compressed conditions should not be interpreted as evidence that compression improves ASR; larger-scale testing is required to determine whether these differences are sampling variation and to identify a genuine degradation threshold.